Dr Ashok Vaid, head of medical oncology at Medanta - The Medicity in Gurugram, and clinical advisor to Indian immunotherapy firm APAC Biotech, "are built around immunotherapy—either to detect a gene to catch a cancer early, or modifying a cell in a way that can help defeat the disease. These new therapies are personalised, differing from case to case as everybody has unique genes and cells." While many immunotherapies are in the research stage, some are entering into clinical trials, and yet others are ready for the market.

HOW DOES IMMUNOTHERAPY WORK?

Immunotherapy classifies tumours as being of two types—hot and cold. Hot tumours such as in melanoma and lung cancer, usually have an accumulation of immune cells (such as proinflammatory cytokines and T cells) that respond well to what is called the immune checkpoint blockade, or ICB, treatment. It involves using drugs to block proteins called 'checkpoints' in the body, which often prevent white blood cells from attacking cancer cells. Cold tumours, such as of the prostate and the pancreas, are those that have a low accumulation of immune cells and, therefore, respond poorly to ICB drugs. Two new revolutionary therapies—CAR-T and NK cell—can alter how the immune cells respond to cancer, CAR-T by genetically altering immune cells to recognise and attack cancer cells, and NK cell by turning cold tumours into hot ones by improving immune cell infiltration.

CAR-T is a revolutionary therapy not only because it offers hope to cancer patients where previously there was none but also because the procedure is one that any hospital can easily include. It involves extracting a cancer patient's white blood cells, similar to how blood is taken at donation camps. These are then sent to a CAR-T company like Immuneel or ImmunoACT, which then reprogrammes the T cells with an antibody-like protein, or the chimeric antigen receptor in CAR. The modified cells are then returned to the patient, in a procedure that takes 10 minutes, and then latch on to the diseased cancer cells and kill them. Scott McIntyre from Illinois, one of the first patients to undergo CAR-T treatment—soon after he was diagnosed with advanced stage lymphoma in 2017—continues to live cancer-free today. The initial diagnosis was that he had only months to live.

Immuneel has tested the therapy on nearly 200 patients in Clinic Barcelona. In India, it has initiated phase 2 clinical trials on patients. Till now, however, the therapy has been so expensive that it could be given only to last-stage cancer patients in India. "Our vision is to make CAR-T affordable and accessible to all," says Purwar. "No one should go without this therapy if it can help them. We are hoping to get the treatment in the market in the next 18 months, and treat many more patients." Shaw concedes it's a challenge but also sees it as an opportunity. "When CAR-T becomes available at an affordable price point, not only can you serve more patients but also provide these treatments at an earlier stage. At an earlier stage, the T cells are far more hardy. The opportunity to engineer hardier T cells gets you better outcomes."



"With immunotherapy, the prognosis for some cancers is so good that you could say that you might die with that cancer but not of it"

KIRAN MAZUMDAR-SHAW
Chairperson, Biocon



MANDAR DEODHAR

Another spin-off of CAR-T, known as CAR-M (chimeric antigen receptor macrophage), is undergoing human trials at the University of Pennsylvania in the US. If successful, it can be used in the treatment of non-haematological cancers as well. "Not just Immuneel, many other companies have recently jumped onto the CAR-T bandwagon and that is very welcome news. I am excited at the thought that we can change the paradigm in India and make this an early stage intervention, almost like a frontline therapy for many haemotological cancers, and get great outcomes," says Shaw.

Another therapy that is much awaited is the NK cell treatment. Like the T cells, NK cells too are white blood cells found naturally in the body. As part of the therapy, such cells are primed to trace and detect early signs of cancer through a process called cytotoxicity, or tumour killing. Cytotoxicity has already been found to be effective in pre-clinical animal